

LOAN PORTFOLIO ANALYTICS & RISK INTELLIGENCE SYSTEM

A Professional Executive Portfolio & Technical Case Study

An Excel-Based Loan Portfolio Analytics, Credit Risk Monitoring & Management Decision-Support Solution

PREPARED BY

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Loan Portfolio Analytics | Credit Risk | Artificial Intelligence | Business Intelligence | Excel Solutions

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PORTFOLIO LINKS

GitHub Repository: [//github.com/PeterKaramwinBanyen/](https://github.com/PeterKaramwinBanyen/)

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2026 • Executive Portfolio

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PROFESSIONAL PROFILE

A multidisciplinary **Data Analytics, Artificial Intelligence and Digital Transformation practitioner** with a growing specialization in **Loan Portfolio Analytics, Credit Risk Monitoring and Financial Data Intelligence**.

My professional practice combines **Data Science, Artificial Intelligence, Machine Learning, Microsoft Fabric, Advanced Microsoft Excel, AI-assisted analytics and prompt engineering** to transform data into actionable business insights and decision-support solutions.

I have developed an **Excel-based Loan Portfolio Analytics & Risk Management System** that applies practical data-analysis techniques to areas such as portfolio performance, loan aging, delinquency, repayment behaviour, risk indicators, portfolio segmentation, concentration analysis and management reporting.

My professional development is supported by certifications and structured learning in **Microsoft Fabric and AI-oriented data solutions, Data Science, and the Principles of Artificial Intelligence**. These provide a strong technical foundation for understanding how modern data and AI technologies can be applied to real-world business and financial challenges.

I am particularly interested in the intersection of **financial data, credit risk, analytics, artificial intelligence and digital transformation**, with a focus on developing practical solutions that improve portfolio visibility, identify emerging risks and support evidence-based management decisions.

PROFESSIONAL FOUNDATION

My professional foundation is built around the integration of **Artificial Intelligence, Data Science, Machine Learning, Financial Data Analytics, Microsoft Excel, Business Intelligence and Digital Transformation**.

1. Artificial Intelligence

My foundation in AI is supported by formal study in the **Principles of Artificial Intelligence**, providing an understanding of fundamental AI concepts and their application to modern business and technology environments.

This foundation supports my interest in:

- AI-assisted data analysis
- Intelligent decision support
- AI agents and Copilot technologies
- Prompt engineering
- AI-enabled business processes
- Practical application of AI to financial and operational challenges

2. Data Science

My Data Science foundation provides the analytical framework for transforming raw data into meaningful information and actionable insights.

Key areas include:

- Data-driven problem solving
- Data preparation and analysis
- Analytical thinking
- Pattern and trend identification
- Business-focused data interpretation
- Data-driven decision support

3. Machine Learning & Microsoft Fabric

My Microsoft learning in **Data Science and Machine Learning using Microsoft Fabric** expands this foundation into modern cloud-based data and AI environments.

This provides exposure to:

- Data science workflows
- Machine learning concepts
- AI-oriented data solutions
- Modern data platforms
- Analytical model development
- Integration of data and AI capabilities

4. Loan Portfolio Analytics & Risk Management

I apply these analytical foundations to the financial-services context through **loan portfolio analytics and risk management**.

Areas of application include:

- Loan portfolio performance
- Portfolio-at-Risk (PAR)
- Non-Performing Loans (NPL)
- Delinquency and arrears
- Loan aging
- Repayment performance
- Risk segmentation
- Portfolio concentration
- Exposure analysis
- Early-warning indicators
- Risk trend analysis
- Management reporting

5. Excel-Based Analytics & Solution Development

Microsoft Excel serves as a practical solution-development environment through which I apply my analytical knowledge to real business problems.

My work includes:

- Structured loan portfolio data models
- Advanced formulas and calculations
- PivotTables and Pivot Charts
- KPI development
- Risk dashboards
- Data validation
- Conditional formatting
- Portfolio segmentation
- Analytical reporting
- Management decision-support tools

6. Digital Transformation

I apply technology not simply as a technical exercise, but as a means of improving business processes, information accessibility, analytical efficiency and decision-making.

My approach is therefore:

Data → Analysis → Insight → Risk Intelligence → Decision Support → Business Action

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

My professional development reflects a deliberate progression across **Artificial Intelligence, Data Science, Machine Learning, Microsoft Fabric and practical business analytics.**

Microsoft 365 & AI / Data Science & Machine Learning

Microsoft — Implement a Data Science and Machine Learning Solution for AI in Microsoft Fabric

Certificate earned: 11 June 2026

Credential ID: peterkaramwinbanyen-5385 | Microsoft Learn

Focus: Data science and machine learning implementation using Microsoft Fabric and AI-oriented data solutions.

Professional relevance: This certification strengthens my understanding of modern data and AI environments and supports the application of analytical and machine-learning concepts to business intelligence and decision-support solutions.

Data Science

IPMC University College — Professional Certificate in Data Science

Certificate earned: 9 July 2026

Credential ID: 0001488

Focus: Data science concepts, analytical thinking, data-driven problem solving, and the application of data techniques to practical business and technology challenges.

Professional relevance: This provides the analytical foundation for working with structured data, identifying patterns and trends, developing meaningful indicators, and converting data into actionable business insights.

Artificial Intelligence

Harvard Online — Certificate of Completion in Principles of AI

Certificate earned: 22 July 2026
Credential ID: 26VY-VBD1-PTN5

Focus: Foundational principles of Artificial Intelligence and the concepts underlying modern AI systems and applications.

Professional relevance: This strengthens my understanding of AI concepts and provides a foundation for exploring AI-assisted analytics, intelligent automation, AI agents and AI-enabled business solutions.

HOW THESE CERTIFICATIONS SUPPORT MY PRACTICE

These certifications form an integrated professional learning pathway that supports my work across **data analytics, artificial intelligence, financial analytics and digital transformation**.

Rather than treating each certification as an isolated achievement, I apply the knowledge progressively:



Applied Professional Practice

The knowledge gained through these certifications is translated into practical solution development.

A key example is my **Excel-based Loan Portfolio Analytics & Risk Management System**, which applies data-analysis principles to loan portfolio monitoring, risk measurement, performance analysis, dashboarding and management reporting.

The project demonstrates the progression from **technical learning to practical business application**:

Learn → Analyze → Design → Develop → Visualize → Monitor → Support Decisions

Continuing Professional Development

My ongoing professional development focuses on strengthening capabilities in:

- Loan Portfolio Analytics
- Credit Risk Analytics
- Financial Data Analysis
- Advanced Microsoft Excel
- Business Intelligence
- Data Visualization
- Artificial Intelligence
- Prompt Engineering
- Machine Learning
- Microsoft Fabric
- AI Agent & Copilot Technologies
- Digital Transformation
- Data Governance and Analytical Controls

This continuous-learning approach enables me to combine **technical knowledge with practical business application**, particularly in environments where financial data, risk management and technology intersect.

Executive Summary

The Loan Portfolio Analytics & Risk Intelligence System was developed as a data-driven solution to digitize lending operations, improve portfolio visibility, automate reporting, and support business decision-making.

The solution integrates:

- Customer Management
- Loan Administration
- Repayment Tracking
- KPI Dashboards
- Portfolio Analytics
- Risk Monitoring
- Executive Reporting

into a unified analytical system.

Business Problem

Many small lending institutions face challenges including:

- Fragmented borrower records
- Manual loan calculations
- Limited portfolio visibility
- Poor risk monitoring
- Delayed reporting

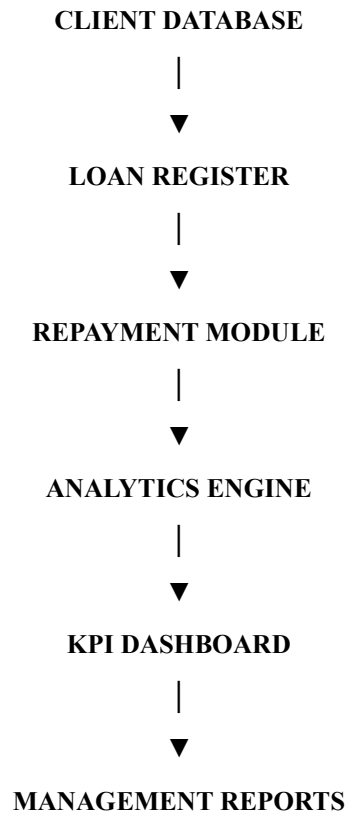
Without structured analytics, management cannot easily monitor loan performance, collection trends, or portfolio exposure.

Project Objectives

The project was developed to:

- ✓ Centralize client information
 - ✓ Digitize loan records
 - ✓ Track repayments
 - ✓ Automate loan calculations
 - ✓ Monitor portfolio risk
 - ✓ Generate management reports
 - ✓ Visualize lending performance through dashboards
 - ✓ Support data-driven decisions
-

System Architecture



DATA SCIENCE METHODOLOGY

Phase 1 – Data Collection

Collected:

- Borrower information
- Loan data
- Repayment transactions

PHASE 2 – DATA CLEANING & STRUCTURING

Created structured datasets:

- Client Database
- Loan Register
- Repayments Table

PHASE 3 – DATA ANALYSIS

Performed:

Portfolio analysis

Repayment analysis

Risk analysis

KPI calculations

PHASE 4 – VISUALIZATION

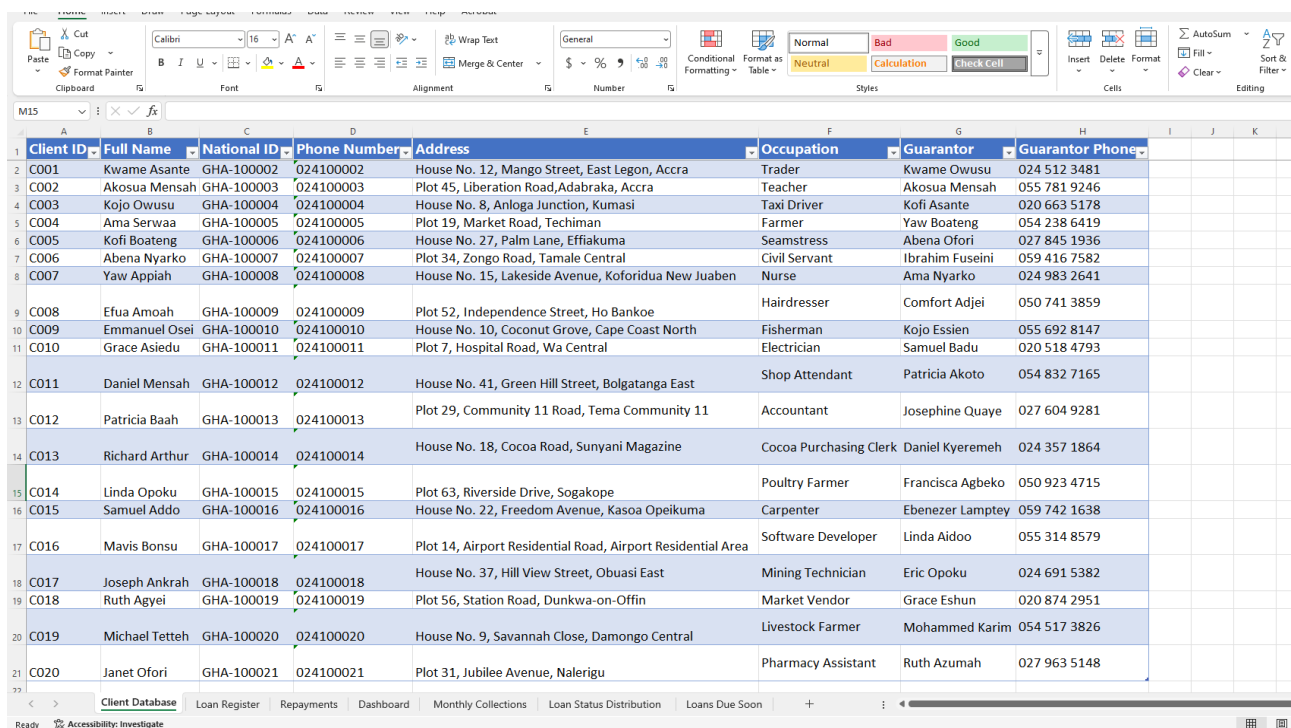
Built:

- Dashboard KPIs
- Collection charts
- Loan status charts

PHASE 5 – BUSINESS INTELLIGENCE

Generated insights and recommendations for decision-makers.

Client Database Module



Client ID	Full Name	National ID	Phone Number	Address	Occupation	Guarantor	Guarantor Phone
C001	Kwame Asante	GHA-100002	024100002	House No. 12, Mango Street, East Legon, Accra	Trader	Kwame Owusu	024 512 3481
C002	Akosua Mensah	GHA-100003	024100003	Plot 45, Liberation Road, Adabraka, Accra	Teacher	Akosua Mensah	055 781 9246
C003	Kojo Owusu	GHA-100004	024100004	House No. 8, Anloga Junction, Kumasi	Taxi Driver	Kofi Asante	020 663 5178
C004	Ama Serwaa	GHA-100005	024100005	Plot 19, Market Road, Techiman	Farmer	Yaw Boateng	054 238 6419
C005	Kofi Boateng	GHA-100006	024100006	House No. 27, Palm Lane, Effiakuma	Seamstress	Abena Ofori	027 845 1936
C006	Abena Nyarko	GHA-100007	024100007	Plot 34, Zongo Road, Tamale Central	Civil Servant	Ibrahim Fuseini	059 416 7582
C007	Yaw Appiah	GHA-100008	024100008	House No. 15, Lakeside Avenue, Koforidua New Juaben	Nurse	Ama Nyarko	024 983 2641
C008	Efua Amoah	GHA-100009	024100009	Plot 52, Independence Street, Ho Bankoe	Hairdresser	Comfort Adjiei	050 741 3859
C009	Emmanuel Osei	GHA-100010	024100010	House No. 10, Coconut Grove, Cape Coast North	Fisherman	Kojo Essien	055 692 8147
C010	Grace Asiedu	GHA-100011	024100011	Plot 7, Hospital Road, Wa Central	Electrician	Samuel Badu	020 518 4793
C011	Daniel Mensah	GHA-100012	024100012	House No. 41, Green Hill Street, Bolgatanga East	Shop Attendant	Patricia Akoto	054 832 7165
C012	Patricia Baah	GHA-100013	024100013	Plot 29, Community 11 Road, Tema Community 11	Accountant	Josephine Quaye	027 604 9281
C013	Richard Arthur	GHA-100014	024100014	House No. 18, Cocoa Road, Sunyani Magazine	Cocoa Purchasing Clerk	Daniel Kyeremeh	024 357 1864
C014	Linda Opoku	GHA-100015	024100015	Plot 63, Riverside Drive, Sogakope	Poultry Farmer	Francisca Agbeko	050 923 4715
C015	Samuel Addo	GHA-100016	024100016	House No. 22, Freedom Avenue, Kasoa Opeikuma	Carpenter	Ebenezer Lamptey	059 742 1638
C016	Mavis Bonsu	GHA-100017	024100017	Plot 14, Airport Residential Road, Airport Residential Area	Software Developer	Linda Aidoo	055 314 8579
C017	Joseph Ankrah	GHA-100018	024100018	House No. 37, Hill View Street, Obuasi East	Mining Technician	Eric Opoku	024 691 5382
C018	Ruth Agyei	GHA-100019	024100019	Plot 56, Station Road, Dunkwa-on-Offin	Market Vendor	Grace Eshun	020 874 2951
C019	Michael Tetteh	GHA-100020	024100020	House No. 9, Savannah Close, Damongo Central	Livestock Farmer	Mohammed Karim	054 517 3826
C020	Janet Ofori	GHA-100021	024100021	Plot 31, Jubilee Avenue, Nalerigu	Pharmacy Assistant	Ruth Azumah	027 963 5148

Figure 1: Client Database Module

Purpose

Stores borrower and guarantor information.

Data Captured

- Client ID
 - Full Name
 - National ID
 - Phone Number
 - Address
 - Occupation
 - Guarantor Details
-

Data Science Value

Demonstrates:

- Master data management
 - Relational design
 - Data structuring
-

Loan Register Module

Loan ID	Client ID	Full Name	National ID	Phone Number	Address	Guarantor Name	Guarantor Phone	Loan Date	Loan Amount (C)	Interest Rate (%)	Loan Term (M)
LN-0001	C001	Kwame Asan	GHA-100002	024100002	House No. 12, Mango Street, East Legon	Kwame Owusu	024 512 3481	15-Jan-25	¢13,500.00	24.00%	
LN-0002	C002	Akosua Men	GHA-100003	024100003	Plot 45, Liberation Road, Adabraka	Ac Akosua Mensah	055 781 9246	28-Jan-25	¢5,000.00	26.50%	
LN-0003	C003	Kojo Owusu	GHA-100004	024100004	House No. 8, Anloga Junction, Kumasi	Kofi Asante	020 663 5178	12-Feb-25	¢8,500.00	28.00%	
LN-0004	C004	Ama Serwaa	GHA-100005	024100005	Plot 19, Market Road, Techiman	Yaw Boateng	054 238 6419	3-Mar-25	¢12,000.00	25.50%	
LN-0005	C005	Kofi Boateng	GHA-100006	024100006	House No. 27, Palm Lane, Effiakuma	Abena Ofori	027 845 1936	21-Mar-25	¢15,000.00	30.00%	
LN-0006	C006	Abena Nyarko	GHA-100007	024100007	Plot 34, Zongo Road, Tamale Central	Ibrahim Fuseini	059 416 7582	14-Apr-25	¢6,500.00	22.50%	
LN-0007	C007	Yaw Appiah	GHA-100008	024100008	House No. 15, Lakeside Avenue, Kofor	Ama Nyarko	024 983 2641	9-May-25	¢2,000.00	27.00%	
LN-0008	C008	Efua Amoah	GHA-100009	024100009	Plot 52, Independence Street, Ho	Ban Comfort Adjeei	050 741 3859	27-May-25	¢10,000.00	31.50%	
LN-0009	C009	Emmanuel C	GHA-100010	024100010	House No. 10, Coconut Grove, Cape C	Kojo Essien	055 692 8147	18-Jun-25	¢4,500.00	29.00%	
LN-0010	C010	Grace Asiedu	GHA-100011	024100011	Plot 7, Hospital Road, Wa Central	Samuel Badu	020 518 4793	11-Jul-25	¢7,800.00	23.00%	
LN-0011	C011	Daniel Mens	GHA-100012	024100012	House No. 41, Green Hill Street, Bolga	Patricia Akoto	054 832 7165	25-Aug-25	¢18,000.00	32.00%	
LN-0012	C012	Patricia Baaf	GHA-100013	024100013	Plot 29, Community 11 Road, Tema	C Josephine Quayee	027 604 9281	16-Sep-25	¢3,500.00	22.00%	
LN-0013	C013	Richard Arth	GHA-100014	024100014	House No. 18, Cocoa Road, Sunyani	N Daniel Kyeremeh	024 357 1864	8-Oct-25	¢25,000.00	26.00%	
LN-0014	C014	Linda Opoku	GHA-100015	024100015	Plot 63, Riverside Drive, Sogakope	Francisca Agbeko	050 923 4715	19-Nov-25	¢14,000.00	28.50%	
LN-0015	C015	Samuel Add	GHA-100016	024100016	House No. 22, Freedom Avenue, Kaso	Ebenezer Lampptey	059 742 1638	15-Dec-25	¢9,000.00	27.50%	
LN-0016	C016	Mavis Bonsu	GHA-100017	024100017	Plot 14, Airport Residential Road, Airp	Linda Aidoo	055 314 8579	20-Jan-26	¢11,500.00	23.50%	
LN-0017	C017	Joseph Ankr	GHA-100018	024100018	House No. 37, Hill View Street, Obuas	Eric Opoku	024 691 5382	17-Feb-26	¢3,000.00	24.50%	
LN-0018	C018	Ruth Agyei	GHA-100019	024100019	Plot 56, Station Road, Dunkwa-on-Off	Grace Eshun	020 874 2951	24-Mar-26	¢22,500.00	33.00%	
LN-0019	C019	Michael Tett	GHA-100020	024100020	House No. 9, Savannah Close, Damon	Mohammed Karim	054 517 3826	13-Apr-26	¢5,800.00	25.00%	
LN-0020	C020	Janet Ofori	GHA-100021	024100021	Plot 31, Jubilee Avenue, Nalerigu	Ruth Azumah	027 963 5148	29-May-26	¢16,000.00	29.50%	

Figure 2: Loan Register Module

Loan Date	Loan Amount (C)	Interest Rate (%)	Loan Term (Months)	Due Date	Expected Monthly Payment	Total Amount to Repay	Amount Repaid	Outstanding Balance	Due Soon	Status	Remarks
15-Jan-25	¢13,500.00	24.00%	6	15-Jul-2025	¢2,255.40	¢13,532.40	¢2,500.00	¢11,032.40	No	Defaulted	
28-Jan-25	¢5,000.00	26.50%	12	28-Jan-2026	¢417.77	¢5,013.25	¢5,100.00	¢86.75	No	Completed	
12-Feb-25	¢8,500.00	28.00%	9	12-Nov-2025	¢947.09	¢8,523.80	¢6,000.00	¢2,523.80	No	Defaulted	
3-Mar-25	¢12,000.00	25.50%	8	3-Nov-2025	¢1,503.83	¢12,030.60	¢12,000.00	¢30.60	No	Defaulted	
21-Mar-25	¢15,000.00	30.00%	10	21-Jan-2026	¢1,504.50	¢15,045.00	¢2,600.00	¢12,445.00	No	Defaulted	
14-Apr-25	¢6,500.00	22.50%	5	14-Sep-2025	¢1,302.93	¢6,514.63	¢15,000.00	¢8,485.38	No	Completed	
8-May-25	¢20,000.00	27.00%	3	9-Aug-2025	¢6,684.67	¢20,054.00	¢7,500.00	¢12,554.00	No	Defaulted	
27-May-25	¢10,000.00	31.50%	2	27-Jul-2025	¢5,015.75	¢10,031.50	¢1,350.00	¢8,681.50	No	Defaulted	
18-Jun-25	¢4,500.00	29.00%	1	18-Jul-2025	¢4,513.05	¢4,513.05	¢3,900.00	¢613.05	No	Defaulted	
11-Jul-25	¢7,800.00	23.00%	4	11-Nov-2025	¢1,954.49	¢7,817.94	¢9,000.00	¢-1,182.06	No	Completed	
25-Aug-25	¢18,000.00	32.00%	7	25-Mar-2026	¢2,579.66	¢18,057.60	¢1,750.00	¢16,307.60	No	Defaulted	
16-Sep-25	¢3,500.00	22.00%	11	16-Aug-2026	¢318.88	¢3,507.70	¢2,000.00	¢16,492.30	No	Completed	
8-Oct-25	¢25,000.00	26.00%	9	8-Jul-2026	¢2,785.00	¢25,065.00	¢8,400.00	¢16,665.00	No	Overdue	
19-Nov-25	¢14,000.00	28.50%	6	19-May-2026	¢2,339.98	¢14,039.90	¢4,500.00	¢9,539.90	No	Defaulted	
15-Dec-25	¢9,000.00	27.50%	3	15-Mar-2026	¢3,008.25	¢9,024.75	¢6,900.00	¢2,124.75	No	Defaulted	
20-Jan-26	¢11,500.00	23.50%	5	20-Jun-2026	¢2,305.41	¢11,527.03	¢24,000.00	¢-12,472.98	No	Completed	
17-Feb-26	¢3,000.00	24.50%	12	17-Feb-2027	¢2,506.13	¢30,073.50	¢11,250.00	¢18,823.50	No	Active	
24-Mar-26	¢22,500.00	33.00%	10	24-Jan-2027	¢2,257.43	¢22,574.25	¢2,320.00	¢20,254.25	No	Active	
13-Apr-26	¢5,800.00	25.00%	11	13-Mar-2027	¢528.59	¢5,814.50	¢9,600.00	¢-3,785.50	No	Completed	
29-May-26	¢16,000.00	29.50%	8	29-Jan-2027	¢2,005.90	¢16,047.20	¢5,400.00	¢10,647.20	No	Active	

Figure 3: Loan Register / Data Capture View

Purpose

Acts as the primary fact table.

Key Metrics Captured

- Principal
- Interest Rate
- Loan Term
- Due Date
- Amount Repaid
- Outstanding Balance
- Loan Status

Business Value

Provides real-time portfolio monitoring and loan performance tracking.

Repayment Analytics Module

Receipt No	Date Paid	Loan ID	Client ID	Amount Paid	Payment Method	Received By	Remarks
RCT0001	14-Feb-2026	LN-0001	C001	2,500.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0002	19-Feb-2026	LN-0002	C002	5,100.00	Bank Transfer	Lender (Melody Esinam Agyeman)	
RCT0003	24-Feb-2026	LN-0003	C003	6,000.00	Cash	Lender (Melody Esinam Agyeman)	
RCT0004	1-Mar-2026	LN-0004	C004	12,000.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0005	6-Mar-2026	LN-0005	C005	2,600.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0006	11-Mar-2026	LN-0006	C006	15,000.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0007	16-Mar-2026	LN-0007	C007	7,500.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0008	21-Mar-2026	LN-0008	C008	1,350.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0009	26-Mar-2026	LN-0009	C009	3,900.00	Bank Transfer	Lender (Melody Esinam Agyeman)	
RCT0010	31-Mar-2026	LN-0010	C010	9,000.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0011	5-Apr-2026	LN-0011	C011	1,750.00	Cash	Lender (Melody Esinam Agyeman)	
RCT0012	10-Apr-2026	LN-0012	C012	20,000.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0013	15-Apr-2026	LN-0013	C013	8,400.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0014	20-Apr-2026	LN-0014	C014	4,500.00	Bank Transfer	Lender (Melody Esinam Agyeman)	
RCT0015	25-Apr-2026	LN-0015	C015	6,900.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0016	30-Apr-2026	LN-0016	C016	24,000.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0017	5-May-2026	LN-0017	C017	11,250.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0018	10-May-2026	LN-0018	C018	2,320.00	Cash	Lender (Melody Esinam Agyeman)	
RCT0019	15-May-2026	LN-0019	C019	9,600.00	Mobile Money	Lender (Melody Esinam Agyeman)	
RCT0020	20-May-2026	LN-0020	C020	5,400.00	Mobile Money	Lender (Melody Esinam Agyeman)	

Figure 4: Repayment Analytics Module

Purpose

Tracks all repayment transactions.

Total Collections

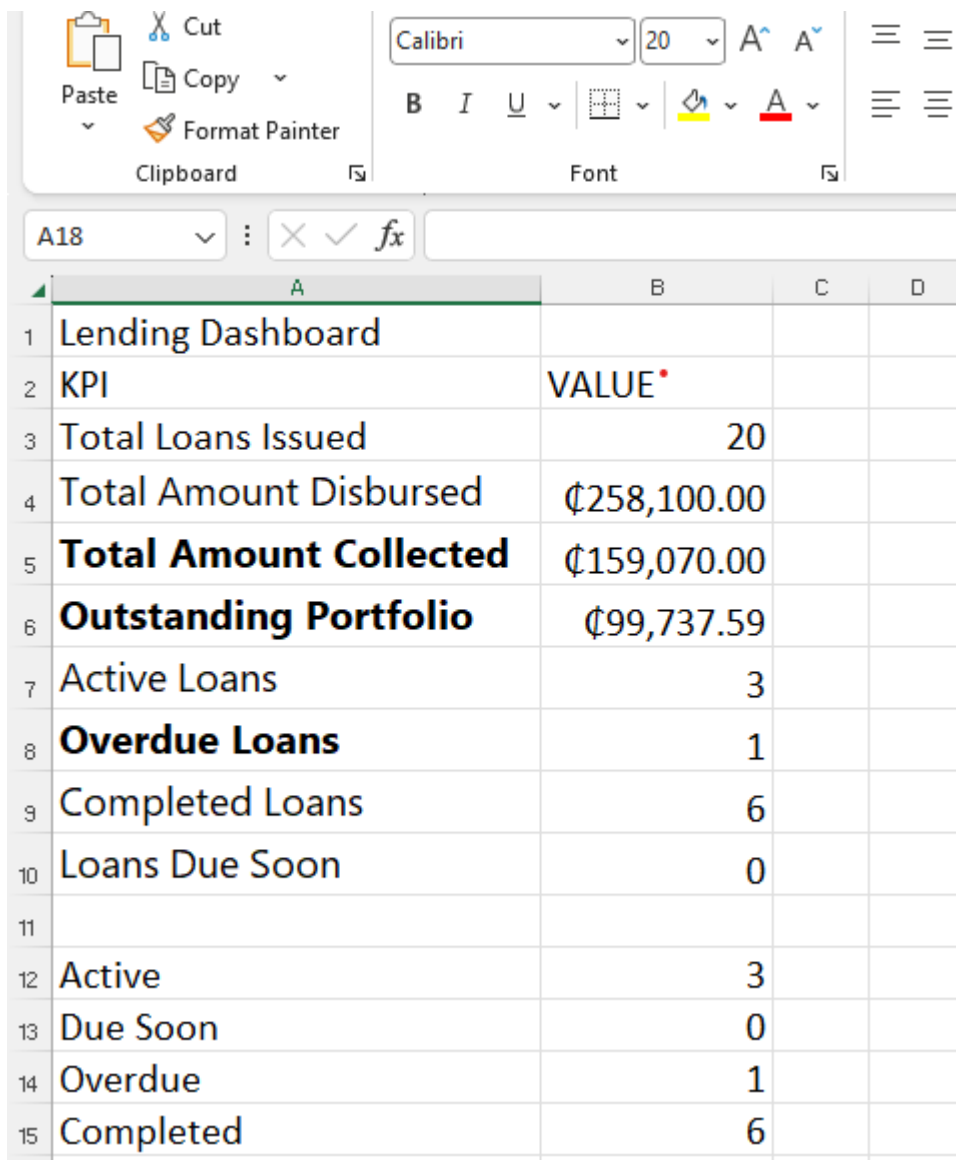
GHS 159,070

Analytical Uses

- Collection analysis
- Payment method analysis
- Transaction auditing

Executive Dashboard

KPI Summary



	A	B	C	D
1	Lending Dashboard			
2	KPI	VALUE*		
3	Total Loans Issued	20		
4	Total Amount Disbursed	¢258,100.00		
5	Total Amount Collected	¢159,070.00		
6	Outstanding Portfolio	¢99,737.59		
7	Active Loans	3		
8	Overdue Loans	1		
9	Completed Loans	6		
10	Loans Due Soon	0		
11				
12	Active	3		
13	Due Soon	0		
14	Overdue	1		
15	Completed	6		

Figure 5: Executive Dashboard – KPI Summary

KPI Analysis

Collection Efficiency

$$159,070 \div 258,100 = 61.63\%$$

Collection Rate: 61.63%

Outstanding Exposure

$$99,737.59 \div 258,100$$

$$= 38.64\%$$

Outstanding Portfolio Ratio:

38.64%

Monthly Collection Trends

Key Finding

Largest repayment:

GHS 24,000 on 30-Apr-2026

Insight

Collection performance accelerated during April and May, indicating stronger repayment activity.

Monthly Collection Efficiency

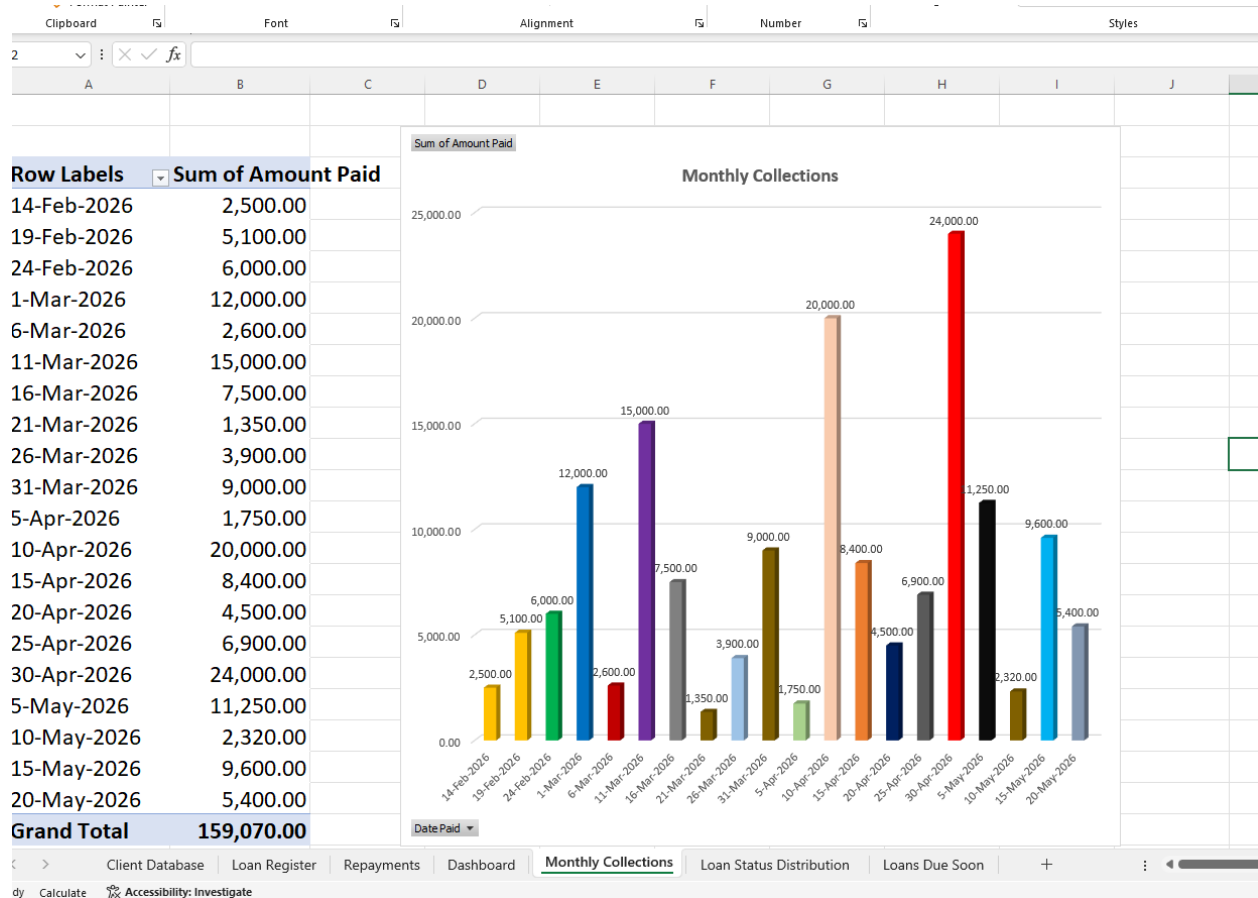


Figure 6: Monthly Collection Efficiency

Loan Status Distribution

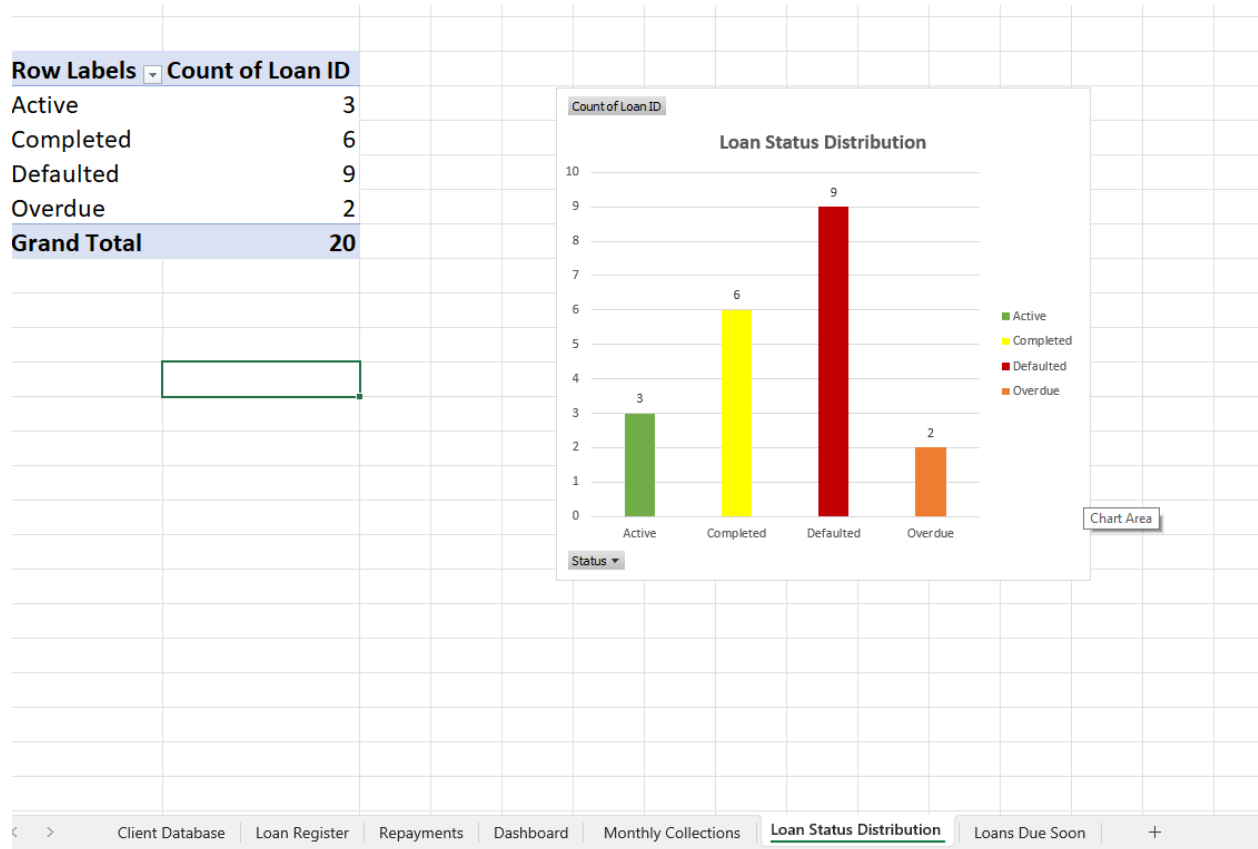


Figure 7: Loan Status Distribution

Observation

45% of loans are defaulted.

$$9 \div 20 = 45\%$$

This represents the most significant portfolio risk.

Portfolio Risk Assessment

High Risk

Defaulted Loans = 9

Medium Risk

Overdue Loans = 2

Exposure

Outstanding Portfolio = **GHS 99,737.59**

Risk Rating Framework

Score	Classification
0–20%	Low
21–40%	Moderate
41–60%	High
>60%	Critical

Portfolio Rating:

 **High Risk (45%)**

Executive Business Insights

Insight 1

The organization has recovered:

GHS 159,070

representing over 61% portfolio recovery.

Insight 2

Defaulted loans represent the largest segment and should be prioritized for collections.

Insight 3

The portfolio still holds nearly:

GHS 99,737.59

in outstanding opportunities.

Insight 4

The dashboard enables management to identify risks without reviewing raw records.

Recommendations Driven by Data

Immediate:

- Implement SMS payment reminders
- Monitor overdue accounts weekly
- Escalate defaulted accounts

Medium-Term

- Portfolio aging analysis
- Borrower risk scoring
- Collection forecasting

Advanced Data Science Roadmap

- Predictive Default Model
 - Credit Risk Scoring Engine
 - Power BI Dashboard
 - Borrower Segmentation
 - Machine Learning Forecasting
-

FORMULAS & LOGIC USED

1. XLOOKUP()
 2. SUMIF()
 3. IF()
 4. EDATE()
 5. Outstanding Balance = Total Amount To Repay - Amount Repaid
 6. Loan Status Classification
-

Skills Demonstrated

Data Science

- Data Analysis
- Financial Analytics
- Insight Generation

Business Intelligence

- KPI Design
- Dashboard Development
- Executive Reporting

Data Management

- Data Modeling
- Data Structuring
- Data Governance

Analytics

- Portfolio Analysis
- Risk Monitoring
- Trend Analysis

Value to Employers

This project demonstrates the ability to:

- Solve business problems using data
- Design analytical systems
- Build dashboards
- Monitor risk
- Automate reporting
- Generate actionable insights

This positions me not merely as an Excel user, but as a **Data Scientist capable of building business intelligence and financial analytics solutions.**

Future Enhancements

Business Intelligence Expansion

- Power BI Dashboard
- Interactive Executive Reporting
- Automated Portfolio Monitoring

Advanced Analytics

- Collection Forecasting
- Credit Risk Scoring
- Borrower Segmentation

Machine Learning Roadmap

- Default Prediction Model
- Probability of Default Scoring
- Early Warning System
- Loan Approval Recommendation Engine

Cloud Transformation

- SQL Database Integration
 - Power BI Service Publishing
 - Real-Time Dashboard Refresh
-

Project Conclusion

The Loan Portfolio Analytics & Risk Intelligence System demonstrates the successful application of Data Science, Business Intelligence, Financial Analytics, Data Modeling, Dashboard Development, and Decision Support principles to a real-world lending environment. The project transformed raw borrower, loan, and repayment data into actionable insights that support portfolio monitoring, risk management, and executive reporting. Through automated calculations, KPI dashboards, and portfolio analytics, the system improves operational efficiency and enables data-driven decision-making.

Portfolio Summary:

- Total Loans Managed: 20
- Total Amount Disbursed: GHS 258,100
- Total Amount Collected: GHS 159,070
- Outstanding Portfolio: GHS 99,737.59
- Collection Efficiency: 61.63%

This project showcases practical competencies in:

- Data Science
 - Financial Analytics
 - Business Intelligence
 - Dashboard Development
 - Data Visualization
 - Data Modeling
 - Reporting Automation
 - Risk Analysis
 - Digitization
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